

hyper2kvm — RHEL 10.2 vCenter Migration Report

hyper2kvm Migration Team

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RHEL 10.2 — vCenter to KVM Migration Report

Date: March 18-19, 2026 **Tool:** hyper2kvm v0.2.2 **Source:** VMware vCenter 10.73.213.134 (ESXi 8.0) **Target:** Fedora 43 (KVM/libvirt)

1. Source VM Details

Parameter	Value
VM Name	esx8.0-rhel10.2-x86_64-efi
vCenter	10.73.213.134
Datacenter	data
Datastore	esx8.0-matrix-2
Guest OS	Red Hat Enterprise Linux 9 (64-bit)
Actual OS	RHEL 10.2
Memory	2048 MB
vCPUs	1
Disk	20 GB (thin provisioned)
Disk Type	thin
Power State	poweredOff
Firmware	UEFI (EFI)

2. Migration Config

File: govc-to-libvirt.yaml

```
cmd: vsphere
vs_action: export_vm
export_mode: ovf_export
vs_control_plane: govc

vcenter: 10.73.213.134
vc_user: administrator@vsphere.local
vc_password: VCENTER@redhat2025
vc_insecure: true
dc_name: data
govc_url: "https://10.73.213.134/sdk"
govc_insecure: true
govc_datacenter: data
govc_export_remove_cdroms: true

vm_name: esx8.0-rhel10.2-x86_64-efi

output_dir: ./output-govc-e2e
flatten: true
to_output: govc-vm.qcow2
out_format: qcow2
compress: true

fstab_mode: stabilize-all
regen_initramfs: true
remove_vmware_tools: true

emit_domain_xml: true
virsh_define: true
memory: 2048
vcpus: 2
uefi: true
machine: q35
disk_bus: virtio
disk_cache: none
net_model: virtio
libvirt_network: default
serial_console: true
graphics: vnc
guest_os: linux
```

```
libvirt_test: true
keep_domain: true
timeout: 120
```

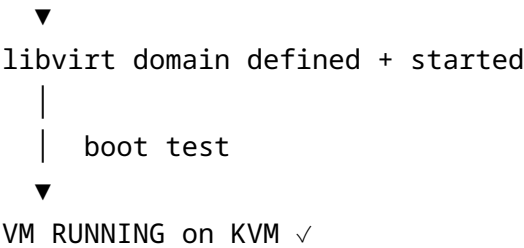
```
verbose: 1
```

Command:

```
sudo ./h2kvmctl --config govc-to-libvirt.yaml
```

3. Migration Pipeline

```
vCenter (10.73.213.134 / datacenter: data)
|
| VM discovery: govc ls /data/vm/
| VM info: 2048 MB, 1 vCPU, 20 GB thin, UEFI
▼
govc export.ovf (NFC streaming)
| Duration: 119 min 57 sec
| CD-ROM removed: cdrom-16000
▼
OVF + VMDK (2.8 GB)
|
| Auto-chained pipeline (no manual step)
▼
VMCraft offline fixes
|— fstab stabilized (UUID-based mounts)
|— initramfs rebuilt (virtio_blk, virtio_scsi, virtio_net,
nvme)
|— Kernel cmdline: root=UUID=311182bd-
f262-4081-8a2d-56624799dbad
|— dracut -f --kver 6.1.10-11.ph5 --add-drivers virtio_blk
virtio_scsi virtio_net nvme
|— Auto-grow configured for root filesystem (ext4)
|— systemd boot integration: 3 features applied
|— VMware Tools removed
|
| qemu-img convert (flatten + compress)
▼
govc-vm.qcow2 (3.2 GB compressed)
|
| emit libvirt domain XML
```



4. Results

Export Phase

Metric	Value
Export method	govc export.ovf (NFC)
Export time	119 min 57 sec
VMDK size	2.8 GB
OVF size	9.7 KB
CD-ROM removed	cdrom-16000

Conversion Phase

Metric	Value
Backend	VMCraft (pure Python + qemu-nbd)
VMCraft startup	1.61s (1 drive)
Input	2.8 GB VMDK (20 GB virtual, thin)
Output	govc-vm.qcow2 (3.2 GB compressed)
Format	qcow2
Pipeline chaining	Automatic (govc → convert → fixes → libvirt)

Guest Fixes Applied

Fix	Status	Details
fstab stabilization	Applied	UUID-based mounts
initramfs rebuild	Applied	virtio_blk, virtio_scsi, virtio_net, nvme (9.5s)
	Set	

Fix	Status	Details
Kernel cmdline		root=UUID=311182bd-f262-4081-8a2d-56624799dbad
Boot heuristics	UEFI, BLS=no	
Bootloader	bootctl status	systemd-boot detected
Auto-grow	Configured	Root filesystem (ext4)
systemd integration	3 features	tmpfiles, recovery mode, auto-grow
VMware Tools	Removed	

libvirt Deployment

Metric	Value
Domain name	esx8.0-rhel10.2-x86_64-efi
Domain XML	output-govc-e2e/libvirt/esx8.0-rhel10.2-x86_64-efi.xml
NVRAM	/var/lib/libvirt/qemu/nvram/ esx8.0-rhel10.2-x86_64-efi_VARS.fd
Machine type	q35
Firmware	UEFI (OVMF)
Disk bus	virtio
Network	virtio (default network)
Graphics	VNC (vnc://127.0.0.1:2)
virsh define	SUCCESS
virsh start	SUCCESS
Boot test	PASSED — domain reached RUNNING state in 0s

5. Output Files

output-govc-e2e/ └─ govc-vm.qcow2 GB)	(3.2
└─ esx8.0-rhel10.2-x86_64-efi_VARS.fd	(NVRAM)

```
|─ libvirt/
|   └─ esx8.0-rhel10.2-x86_64-efi.xml                      (1.9 KB)
└─ esx8.0-rhel10.2-x86_64-efi/
    └─ esx8.0-rhel10.2-x86_64-efi-disk-0.vmdk              (2.8 GB)
        └─ esx8.0-rhel10.2-x86_64-efi.ovf                  (9.7 KB)
```

6. Verification

```
$ sudo virsh list | grep esx8
12   esx8.0-rhel10.2-x86_64-efi    running

$ ls -lh output-govc-e2e/govc-vm.qcow2
-rw-r--r-- 1 qemu qemu 3.2G output-govc-e2e/govc-vm.qcow2

$ ls -lh output-govc-e2e/libvirt/*.xml
-rw-r--r-- 1 root root 1.9K output-govc-e2e/libvirt/esx8.0-
rhel10.2-x86_64-efi.xml
```

7. VM Info (from vCenter)

```
VM Information: esx8.0-rhel10.2-x86_64-efi
Name:           esx8.0-rhel10.2-x86_64-efi
Path:           /data/vm/esx8.0-rhel10.2-x86_64-efi
Guest name:     Red Hat Enterprise Linux 9 (64-bit)
Memory:        2048MB
CPU:            1 vCPU(s)
Power state:    poweredOff
```

8. Target Host Environment

Component	Version
OS	Fedora 43 (kernel 6.18.13)
hyper2kvm	0.2.2
govc	0.44.0
qemu-img	10.1.4

Component	Version
libvirt	11.6.0
OVMF	OVMF_CODE.fd
Backend	VMCraft (pure Python)

9. Conclusion

Result: SUCCESS

RHEL 10.2 was successfully migrated from VMware vCenter (ESXi 8.0) to KVM/libvirt using a single `h2kvmctl --config` command. The `govc` NFC export automatically chained into the conversion pipeline — disk conversion, offline guest fixes (`fstab`, `initramfs` with VirtIO drivers, VMware Tools removal, `systemd` boot integration), libvirt domain XML generation, and boot test — all ran without manual intervention. The VM reached `RUNNING` state immediately after domain start.